

Department Overview

- **Department Name:** Department of Electrical Engineering
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Establishment:** Founded in 2012. The graduate (MS) program was subsequently initiated in Spring 2014.
- **Accreditation:** The undergraduate program is accredited by the **Pakistan Engineering Council (PEC) under Level-II**, indicating compliance with high-quality education standards.
- **Core Focus:** Focuses heavily on Electric Power Systems Engineering and Telecommunication Engineering. The curriculum balances mathematical depth and engineering fundamentals with professional skills like personality grooming, public communication, entrepreneurship, and management.

Vision and Mission

- **Vision:** To become a global leader in the field of Electrical Engineering having a pivotal role in the development and prosperity of society.
- **Mission:** To impart quality learning in the field of electrical engineering, promote excellence in research & innovation, and play a leading role in economic, social, and cultural development.

Program Educational Objectives (PEOs)

The department's educational goals are designed to produce professional engineers who will:

1. Demonstrate professional competence in electrical engineering through in-depth knowledge and skills, alongside contemporary interdisciplinary studies.
2. Engage in continual professional development through effective communication, teamwork, and lifelong learning.
3. Depict professional integrity, practice, and implement ethics and social responsibility to the global community.

Bachelor of Science (BS) Program

Admission Requirements

- **F.Sc (Pre-Engineering):** Completed from any recognized board or equivalent with at least **60% marks**.
- **Diploma of Associate Engineer (DAE):** Completed in a relevant engineering field with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 136
- **Semester Credit Load:** 15–18 credit hours per semester.
- **Minimum Academic Standing:** CGPA $\geq$ 2.0
- **Specialization Streams:** Students can choose to specialize in either Telecommunication Engineering or Power Systems Engineering.

Curriculum Plan (First 4 Semesters Schema)

Note: Credit hour listings follow the format (Theory + Lab).

Semester	S. No.	Course Code	Subject / Course Title	Knowledge Area	Credit Hours
1st	1	HUM-163	Functional English	Humanities	3+0
	2	MATHP-118	Calculus and Analytical Geometry	Natural Sciences	3+0
	3	EE-112/L	Linear Circuit Analysis & Lab	Engineering Foundation	3+1
	4	CS-123	Applications of ICT	Computer Sciences	2+1
	5	ENGG-217L	Engineering Drawing	Engineering Foundation	0+1
	6	PHY-205/L	Applied Physics & Lab	Natural Sciences	2+1
	7	MTH-XX	*Math Deficiency – I (if applicable)	Natural Sciences	3+0*
			Semester Total Credit Hours		17

2nd	1	MATHA-214	Differential Equations	Natural Sciences	3+0
	2	XX	Natural Science Elective	Natural Sciences	3+0
	3	ELE-102L	Electrical Workshop Practice	Engineering Foundation	0+1
	4	HUM-XX	Pakistan Studies	Humanities	2+0
	5	CS-112/L	Computer Programming & Lab	Advanced Computer & IT	3+1
	6	EE-208/L	Electronic Devices & Circuits & Lab	Engineering Foundation	3+1
	7	MTH-XX	*Math Deficiency – II (if applicable)	Natural Sciences	3+0*
			Semester Total Credit Hours		17
3rd	1	MATHA-213	Complex Variables and Transforms	Natural Sciences	3+0

	2	HUM-116	Ideology and Constitution of Pakistan	Humanities	2+0
	3	EE-110/L	Digital Logic Design & Lab	Engineering Foundation	3+1
	4	CS214/L	Data Structures and Algorithm & Lab	Advanced Computer & IT	3+1
	5	ELE-201/L	Electrical Network Analysis & Lab	Engineering Foundation	3+1
	6	ENVIRON-103	Occupational Health and Safety	Multidisciplinary Eng.	1+0
			Semester Total Credit Hours		18
4th	1	HUM-XX	Social Sciences Elective	Humanities	2+0
	2	MATHP-111	Linear Algebra	Natural Sciences	3+0
	3	HUM-101	Islamic	Humanities	2+0

			Studies/Ethics		
	4	MATHA-217	Probability Methods in Engineering	Engineering Foundation	3+0
	5	EE-261/L	Signals and Systems & Lab	Engineering Foundation	3+1
	6	EE-111	Electromagnetic Field Theory	Engineering Foundation	3+0
			Semester Total Credit Hours		17

Core Higher Semester Subjects (Semesters 5 to 8 Highlights)

- **5th Semester:** Electrical Machines (3+1), Microprocessors and Interfacing (3+1), Communication Systems (3+1), Civics and Community Engagement (2+0), MDEE Elective (3+0). Total: 17 Credits.
- **6th Semester:** Expository Writing (3+0), Linear Control Systems (3+1), Power Distribution and Utilization (3+1), Depth Elective (Core)-1 (3+1), Project Management (2+0). Total: 17 Credits.
- **7th Semester:** Depth Elective–II Core (3+1), Depth Elective–III (3+1), Flexible Elective I (**Climate Change and Risk Management**, 2+1), Depth Elective–IV (3+0), Arts & Humanities Elective (**Muslim Heritage**, 2+0), Final Year Design Project Part-I (0+2). Total: 18 Credits.
- **8th Semester:** Entrepreneurship (2+0), Depth Elective–V (3+0), Flexible Elective–II (3+0), Flexible Elective–III (3+0), Final Year Design Project Part-II (0+4). Total: 15 Credits.

Master of Science (MS) Program

Admission Requirements

- **Academic Background:** 16 years of education or equivalent (such as a 4-year B.E or BS

degree) in a relevant field from an HEC-recognized institution.

- **Minimum Grades:** At least **60% marks** under the annual system, or a minimum CGPA of **2.5 out of 4.0**.
- **Testing:** A valid **GAT General** score with at least **50% marks**.

Degree Requirements

- **Total Credit Hours:** 30
- **Coursework:** 24 credit hours of graduate-level courses (8 courses).
- **Research Thesis:** 6 credit hours of research conducted under the supervision of PhD faculty members.
- **Minimum Passing Standing:** CGPA $\geq$ 2.5
- **Areas of Research:** Specialized streams are available in **Telecommunication Systems Engineering** and **Power Systems Engineering**.

Future Program Plans

The department is actively working toward launching a **Ph.D. Engineering Program**, which will broaden specialized research tracks across several fields:

- Power Systems Engineering
- Telecommunication Systems Engineering
- Electronic Engineering
- Control Systems Engineering
- Computer Systems Engineering